

ST GEORGE ISLAND, AK, USA

WMO: 703835

Lat: 56.600N

Lon: 169.565W

Elev: 125

StdP: 14.63

Time Zone: -9.00 (NAK)

Period: 98-19

WBAN: 25628

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	6.9	9.9	1.0	5.8	8.1	3.8	6.7	10.9	47.2	24.1	44.0	28.3	17.8	340	0.990

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.1	55.3	52.6	54.1	51.9	52.6	50.9	53.5	54.5	52.5	53.4	51.4	52.2	13.8	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
52.9	59.8	54.0	51.9	57.7	53.0	50.6	54.9	51.8	22.4	54.5	21.7	53.3	21.2	52.3	61.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 42.0	37.0	33.7	DB	5.7	60.0	6.3	4.5	1.1	63.2	-2.6	65.9	-6.2	68.4	-10.8	71.7	(4)
(5)			WB	5.6	55.1	5.8	2.7	1.5	57.0	-1.9	58.7	-5.2	60.2	-9.4	62.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.0	27.7	29.1	27.6	31.9	37.0	43.1	47.7	49.4	46.2	39.9	34.2	30.2
	DBStd	9.44	7.53	7.93	7.85	5.69	4.35	3.06	3.09	2.70	3.39	4.31	4.81	6.42
	HDD50	4773	693	585	694	543	402	209	84	44	118	314	473	614
	HDD65	10203	1158	1005	1159	993	867	658	536	483	563	779	923	1079
	CDD50	42	0	0	0	0	0	0	13	25	4	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	16.6	21.1	20.5	18.7	17.5	15.0	12.7	11.2	12.1	15.1	17.4	19.6	19.2
Precipitation	PrecAvg	41.7	3.8	3.7	3.0	2.0	2.3	2.7	3.9	3.8	4.3	3.5	4.4	4.3
	PrecMax	49.0	6.4	7.5	6.4	3.7	4.0	4.2	5.9	5.3	7.2	6.1	7.0	6.8
	PrecMin	32.8	1.6	1.3	1.0	0.9	1.0	0.9	1.6	1.5	2.3	1.9	1.9	2.4
	PrecStd	3.9	1.1	1.3	1.3	0.6	0.7	0.9	1.1	0.8	1.1	0.9	1.3	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.4	41.2	40.5	43.2	49.3	54.6	59.4	58.2	55.0	49.5	45.3	43.0
		MCWB	39.1	40.4	38.6	39.9	44.2	49.0	53.2	54.6	52.8	48.2	43.6	41.4
	2%	DB	38.8	39.4	39.1	41.4	46.1	50.9	55.2	55.6	53.2	48.3	43.8	40.8
		MCWB	37.6	38.2	37.6	39.4	42.6	47.3	52.0	53.1	51.0	46.9	42.2	39.6
	5%	DB	37.3	38.2	37.4	39.8	44.5	48.8	53.5	54.4	51.9	47.1	42.8	39.3
		MCWB	36.2	37.0	36.0	38.1	41.7	46.4	51.3	52.5	50.1	45.7	41.3	38.0
	10%	DB	36.4	37.1	36.4	38.7	42.9	47.4	52.0	53.4	50.6	45.8	41.1	38.0
		MCWB	35.4	35.9	35.0	37.2	40.8	45.4	50.4	51.9	48.9	44.1	39.9	36.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.6	40.9	39.8	41.4	45.5	50.1	54.9	55.5	53.6	49.1	44.6	42.1
		MCDB	40.0	41.0	40.6	42.1	48.3	53.0	57.5	57.2	54.1	49.3	44.9	42.6
	2%	WB	38.0	38.7	38.1	39.8	43.5	48.1	53.0	54.1	51.7	47.7	43.1	40.1
		MCDB	38.5	39.1	38.7	40.9	45.1	50.1	54.5	55.1	52.3	48.1	43.7	40.7
	5%	WB	36.8	37.4	36.7	38.6	42.2	46.6	51.7	53.1	50.6	46.2	41.6	38.5
		MCDB	37.4	37.9	37.4	39.7	43.9	48.2	52.8	53.9	51.3	46.7	42.4	39.1
	10%	WB	35.6	36.3	35.2	37.5	41.1	45.9	50.5	52.1	49.6	44.9	40.0	37.1
		MCDB	36.2	37.0	36.1	38.5	42.9	47.2	51.5	52.9	50.4	45.7	41.2	37.8
Mean Daily Temperature Range	5% DB	MDBR	6.1	6.6	6.7	6.6	6.7	6.5	5.5	5.1	5.8	6.2	6.0	6.8
		MCDBR	4.8	5.4	6.4	6.7	8.9	9.1	8.2	6.8	6.5	5.2	6.3	6.4
		MCWBR	4.9	6.0	6.0	5.8	6.4	6.3	5.9	5.7	5.3	5.0	6.6	6.5
	5% WB	MCDBR	4.8	5.7	6.2	6.3	7.9	8.1	7.2	5.9	5.2	4.4	6.1	5.9
		MCWBR	5.1	6.4	6.1	5.6	6.0	6.0	5.6	5.3	4.8	5.0	6.6	6.5
Clear-Sky Solar Irradiance	taub	0.302	0.340	0.335	0.399	0.405	0.389	0.395	0.388	0.370	0.340	0.324	0.274	
	taud	2.226	2.234	2.333	2.138	2.217	2.366	2.390	2.412	2.365	2.317	2.167	2.340	
	Ebn at Noon	182	222	261	257	264	271	266	260	246	220	165	171	
	Edh at Noon	17	25	30	42	41	36	34	32	29	23	18	12	
All-Sky Solar Radiation	RadAvg	133	321	725	1119	1321	1341	1159	980	768	459	176	93	
	RadStd	15	23	114	117	98	90	92	89	54	24	17	12	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page